

Lab 5: Labor Transformation, Automation & Displacement

Nishanth Ravula · PhD 832: Ethics in Responsible AI · September 2026

Tools: TripleTen “Will AI take my job?” calculator; AI Labor Index; OECD AI Incidents and Hazards Monitor (AIM)

Case: Duolingo contractor reduction (end of 2023, reported January 2024) · Framework: Ponce Del Castillo & Taes (2025), Chapter 17 in Smuha (Ed.)

Main finding. The two tools rank my three occupations in the same order (Software Engineer > Business Analyst > Medical Researcher), but they disagree sharply about how big the gaps are. The difference comes from what each tool measures. The AI Labor Index measures what AI *could* do to a job’s tasks. TripleTen blends in how much AI people are *actually* using. None of the three roles looks likely to disappear. All three look set to be restructured, with some tasks sped up or taken over by AI while the parts that need judgment and accountability stay with people.

The Duolingo case shows that the same kind of exposure (AI that generates text and translations) can end in very different outcomes for workers. It depends on what the organization decides to do with the productivity gain. That is why I argue that task automation becomes a labor-governance problem at the point where a gain in speed is turned into less work, less pay, or a lower-skilled role, and the affected workers have no say in that decision.

1. Step 1: Comparative Labor Risk Analysis

1.1 Occupations, inputs, and matching

I ran three occupations through TripleTen and matched each one to the closest AI Labor Index occupation. TripleTen personalizes its score using the inputs below, so I recorded them. The AI Labor Index does not take personal inputs. It scores the occupation as a whole, based on its O*NET task list. (O*NET is the U.S. Department of Labor’s database of occupations and the tasks each one involves.)

TripleTen job title	Industry	Company size	Experience	Remote	AI Labor Index match	Match quality
Software Engineer	Technology & Engineering	Small (1–50)	Senior (5+ yrs)	No	Software Engineer	Exact
Business Analyst	Finance & Accounting	Enterprise (500+)	Mid (3–5 yrs)	Yes	Business & Management Consultant	Close. The Index groups Management Analysts (the O*NET home of business analysts) with three related sub-occupations.
Medical Researcher	Healthcare & Medical	Medium (51–500)	Mid (3–5 yrs)	No	Medical Researcher	Exact

Controlled baseline. My original inputs were not the same across the three roles, so part of the TripleTen gap could reflect the profiles I entered rather than the occupations. To test this, I reran Software Engineer and Business Analyst with the profile already used for Medical Researcher: Medium company (51–500), Mid experience (3–5 years), not remote. Each role kept its own industry.

Role	Original profile	Original score	Controlled profile	Controlled score
Software Engineer	Small, Senior, not remote	45%	Medium, Mid, not remote	53% (+8)
Business Analyst	Enterprise, Mid, remote	29%	Medium, Mid, not remote	25% (-4)
Medical Researcher	Medium, Mid, not remote	19%	Same (no rerun needed)	19% (0)
<i>Gap: Software Eng. vs. Business Analyst</i>		16 points		28 points
<i>Gap: Business Analyst vs. Medical Res.</i>		10 points		6 points
<i>Gap: Software Eng. vs. Medical Res.</i>		26 points		34 points

Table 1. TripleTen controlled baseline (TripleTen, n.d.). Screenshots in Appendices A1 and A2.

With the profile held constant, the ranking does not change (Software Engineer > Business Analyst > Medical Researcher), but the spacing does. The profiles I first entered pushed the scores *toward* each other: the senior profile lowered Software Engineer by 8 points, and the enterprise remote profile raised Business Analyst by 4. Controlled, the total spread grows from 26 to 34 points, and most of it now sits between Software Engineer and the other two roles. So the occupation (with its industry) drives the differences, and my original inputs understated them rather than inflating them.

The explanation text shows which inputs mattered. The senior Software Engineer run said “five years of context” is “the part that transfers worst to a tool, and it pulls your number down”; the mid-level run dropped that sentence. The enterprise Business Analyst run said large companies run “process-heavy work, and that is what automation budgets go looking for”; the medium company run replaced this with a note about finance’s long history of automating documented processes. In the tool’s own words, experience lowers exposure and company size raises it. That supports my reading that AI pressure falls hardest on less experienced workers in large organizations. Because each rerun changed more than one input at a time, it cannot say exactly how many points come from each. The AI Labor Index numbers are not affected, because the Index takes no personal inputs.

1.2 Automation vs. AI Exposure Table

Occupation	TripleTen risk result	AI Labor Index Automation / LLM & AI	Human Complexity	AI Adoption	My interpretation
Software Engineer	45% , “Moderate exposure.” Writing code in known patterns is exposed. Deciding what to build, spotting bad designs, and owning systems in production are not. BLS projects about 16% job growth through 2034.	Automation 3.10/5 LLM & AI 16.2% Software 46.7% Robotics & IoT 0.4%	3.71/5	28.4%	Augmentation with restructuring. It has the highest exposure on every measure and by far the highest real-world adoption, yet employment is still projected to grow. AI is already <i>speeding up</i> coding tasks rather than removing the occupation. The pressure lands on routine coding, the kind of work early-career engineers usually do. With a mid-level profile the score rises to 53% (Table 1). Judgment and ownership become a bigger share of the job.

Occupation	TripleTen risk result	AI Labor Index Automation / LLM & AI	Human Complexity	AI Adoption	My interpretation
Business Analyst (Index: Business & Management Consultant)	29% , “Lower exposure.” Reports, planning documents, and status-chasing are what AI handles well. Deciding, prioritizing, and being accountable for outcomes are not. The occupation is projected to grow.	Automation 3.09/5 LLM & AI 15.2% Software 36.4% Robotics & IoT 0.0%	4.03/5	1.3%	Partial task automation leading to restructuring. Its Automation score is almost identical to Software Engineer’s (3.09 vs. 3.10), but adoption is about 22 times lower. With a controlled profile its TripleTen score falls to 25% (Table 1), which widens its distance from Software Engineer even though the two roles are almost equally automatable on the Index. The task list (Section 1.4) shows the exposed work is document production and information gathering. The planning, regulatory, and strategy tasks carry high complexity and accountability. The role is likely to be transformed, not eliminated.
Medical Researcher	19% , “Lower exposure.” Licensing, physical presence, and legal responsibility keep the work with people. AI arrives as an instrument (imaging, notes, decision support). Regulation slows change.	Automation 2.90/5 LLM & AI 11.1% Software 34.6% Robotics & IoT 7.4%	4.09/5	0.3%	Augmentation. It has the lowest exposure and highest complexity of the three, and almost no observed adoption. It is the only role with meaningful robotics exposure (7.4%), which points to lab automation rather than text generation. AI is likely to change the tools researchers use, not whether the job exists.

Table 2. Side-by-side results. TripleTen figures are its “task exposure” score, which it describes as “not a probability of losing your job.” AI Labor Index Automation and Complexity are 1–5 averages across O*NET tasks. LLM & AI, Software, and Robotics & IoT are the Index’s average automation scores by technology type. Adoption is observed in AI use. Sources: TripleTen (n.d.); AI Labor Index (n.d.-a, n.d.-b, n.d.-c). Screenshots are in the Appendix.

1.3 Side-by-side calculations

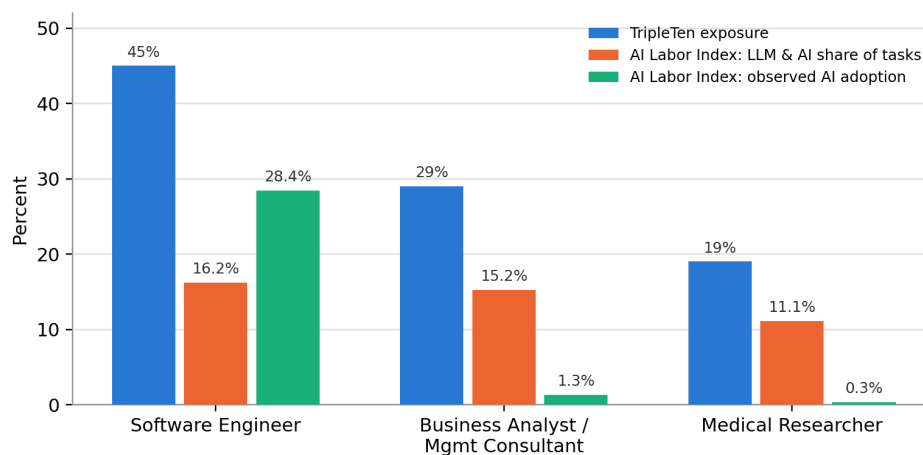
Because the tools use different scales, I did not convert one into the other. Instead I compared (a) whether the tools agree on the *ranking*, (b) how *spread out* each measure is across the three roles (highest value divided by lowest), and (c) three derived measures.

Measure	Software Eng.	Business Analyst	Medical Res.	Spread (max ÷ min)	What it shows
TripleTen exposure (original profiles)	45%	29%	19%	2.37×	Wide spread
TripleTen, controlled profile (Table 1)	53%	25%	19%	2.79×	Wider spread once profiles are equal
AI Labor Index Automation (1–5)	3.10	3.09	2.90	1.07×	Almost flat: 0.20 points on a 5-point scale
LLM & AI share	16.2%	15.2%	11.1%	1.46×	Moderate spread

Measure	Software Eng.	Business Analyst	Medical Res.	Spread (max ÷ min)	What it shows
Observed AI adoption	28.4%	1.3%	0.3%	94.7×	Extreme spread
Human Complexity (1–5)	3.71	4.03	4.09	1.10×	Reverse order: less exposed roles are more complex
<i>Derived:</i> TripleTen minus LLM & AI share (percentage points)	28.8	13.8	7.9	—	TripleTen runs furthest above the Index where adoption is highest
<i>Derived:</i> Complexity buffer (Complexity minus Automation)	0.61	0.94	1.19	1.95×	The more a role depends on complex judgment, the less of it AI can reach
<i>Derived:</i> Adoption ÷ LLM & AI share	1.75	0.086	0.027	65×	How far real use has caught up with technical exposure

Table 3. Comparison calculations. All exposure measures agree on the ranking (Software Engineer > Business Analyst > Medical Researcher). Human Complexity ranks the roles in exactly the reverse order.

Panel A. Percentage measures



Panel B. 1–5 scale measures

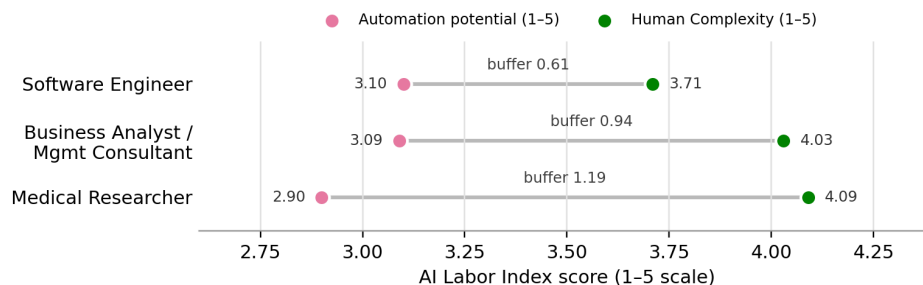


Figure 1. Panel A compares TripleTen exposure with two AI Labor Index percentage measures. They measure different things, so compare the pattern across roles, not bar heights between measures. TripleTen values here use the original profiles; controlled values are in Table 1. Panel B plots the two 1–5 scale measures. The gap between Human Complexity and Automation potential (the “complexity buffer”)

widens as exposure falls: 0.61 for Software Engineer, 0.94 for Business Analyst, 1.19 for Medical Researcher.

Interpretation: probabilistic automation vs. task acceleration. A *probabilistic automation* reading treats a score as the chance that a machine can do the work instead of a person, which would imply substitution. A *task acceleration* reading treats it as how much of the work AI can speed up while a person still does it, which implies a complement. Neither tool’s score, on its own, tells us which of these will happen. The calculations point to three conclusions:

- **Exposure is similar; adoption is not.** On the AI Labor Index, Software Engineer and Business Analyst are technically almost equally automatable (3.10 vs. 3.09). Their realized AI use differs by a factor of 22. The main difference between these roles right now is how organizations have chosen to deploy AI, not what AI can do.
- **TripleTen follows adoption.** TripleTen says its score “combines measured AI use by occupation with US Bureau of Labor Statistics employment projections.” That explains why it sits 28.8 points above the Index’s LLM share for Software Engineer but only 7.9 points above for Medical Researcher. TripleTen is partly measuring where AI has already arrived. Although the lab describes it as the older occupation-level model, its own disclaimer shows it has moved toward task exposure plus observed use.
- **Complexity acts as a buffer.** Human Complexity rises exactly as exposure falls. Measured as Complexity minus Automation, the buffer nearly doubles from Software Engineer (0.61) to Medical Researcher (1.19) (Figure 1, Panel B). Roles built on high-judgment work keep a wider margin that current AI does not reach, which is one reason their observed adoption is so low.
- **High exposure alongside job growth means acceleration, not replacement.** Software Engineer has the highest exposure *and* projected employment growth of about 16%. That pattern fits task acceleration within a growing occupation. The AI Labor Index itself cautions that automation potential does not translate directly into lost jobs or wages.

1.4 Task-level analysis: Business & Management Consultant

The Index lists 66 tasks for this occupation (AI Labor Index, n.d.-a) across four sub-occupations: Business Continuity Planners (BCP), Operations Research Analysts (ORA), General and Operations Managers (GOM), and Management Analysts (MA). I recorded the 19 highest-ranked tasks, sorted by Automation score. The full scores are below. All tasks on the first page are labeled “Digital” in task nature.

Task (abridged)	Sub-occ.	Complexity	Automation	LLM & AI	Accountability
Establish, maintain, or test call trees for disaster communication	BCP	3	4	3	3
Prepare reports summarizing operational or financial results	BCP	3	4	4	3
Analyze corporate intelligence data to identify threats	BCP	4	4	4	4
Study and analyze alternative courses of action	ORA	4	4	4	3
Develop time and cost networks to plan and control projects	ORA	4	4	3	3
Prepare staff work schedules and assign duties	GOM	3	4	3	2
Review financial statements and performance data	GOM	4	4	3	3

Task (abridged)	Sub-occ.	Complexity	Automation	LLM & AI	Accountability
Gather and organize information on problems or procedures	MA	3	4	4	3
Develop emergency management plans for recovery and continuity	BCP	5	3	3	4
Develop disaster recovery plans for physical sites (e.g., data centers)	BCP	4	3	3	4
Write reports summarizing testing activities and recommendations	BCP	3	3	4	3
Analyze impact and risk to essential business functions	BCP	4	3	3	3
Review existing disaster recovery and continuity plans	BCP	4	3	3	3
Conduct or oversee contingency plan integration and operation	BCP	4	3	3	4
Interpret government regulations and codes to ensure compliance	BCP	5	3	3	4
Create scenarios to reestablish operations after disruptions	BCP	5	3	3	4
Create or administer training and awareness materials	BCP	3	3	4	3
Identify strategic improvement and risk-mitigation opportunities	BCP	5	3	3	4
Create business continuity and disaster recovery budgets	BCP	4	3	3	3

Table 4. Task scores (out of 5), 19 of 66 tasks. Data from AI Labor Index (n.d.-a). Bold marks the highest LLM, complexity, and accountability ratings.

Group of tasks	n	Mean Complexity	Mean LLM & AI	Mean Accountability
Automation = 4/5	8	3.50	3.50	3.00
Automation = 3/5	11	4.18	3.18	3.55
Accountability = 4/5	7	4.57	3.00	4.00

Table 5. Group averages calculated from Table 4.

Task set	n	Mean Automation	Mean Complexity	Source
Top 19 tasks (analyzed above)	19	3.42	3.89	Calculated from Table 4
All 66 tasks	66	3.09	4.03	AI Labor Index occupation scores
Remaining 47 tasks	47	≈2.96	≈4.08	Derived: $(66 \times \text{all-task mean} - 19 \times \text{top-19 mean}) \div 47$

Table 6. Checking for selection bias. The derivation assumes the Index's occupation scores are simple averages of its task scores. Because the published scores are rounded to two decimals, the derived values are accurate to about ±0.01.

Selection bias check. I analyzed the top 19 tasks because the Index lists tasks sorted by Automation. That subset overstates the occupation's exposure, so I compared it with the rest (Table 6). The top 19 average 3.42 on Automation. The remaining 47 average about 2.96, which is below the occupation-wide 3.09 and consistent with the sort order, since no remaining task can score above 3. The remaining tasks are also

more complex (about 4.08 vs. 3.89). The full task list therefore makes the case for transformation over elimination *stronger* than the subset alone does.

- **Most susceptible to AI.** The eight Automation-4 tasks are about summarizing, gathering, scheduling, and analyzing data. The six tasks with the highest LLM score (4/5) are nearly all *text production*: operational reports, testing reports, information gathering, and training materials. This is the same type of work Duolingo moved to generative AI.
- **Needs human judgment and accountability.** All four complexity-5 tasks (emergency plans, interpreting regulations, recovery scenarios, and strategic risk mitigation) score only 3 on Automation and 4 on Accountability. Across the seven accountability-4 tasks, average complexity is 4.57. When something goes wrong in these tasks, a named person answers for it.
- **Transformed or eliminated?** Transformed. The most automatable tasks are less complex (3.50 vs. 4.18) and carry less accountability (3.00 vs. 3.55) than the rest. The likely change is that AI drafts the report while the analyst decides what the report means and is responsible for the plan. The risk is uneven, though. Staff whose jobs are mostly the Automation-4 tasks, often junior analysts, are exposed in a way the occupation-level average hides.

2. Step 2: Forensic Incident Audit (OECD AIM)

2.1 The OECD record

[Please see methodology and disclosures here.](#)

Enter a concept or a keyword and press enter. Use advanced options to further narrow the search. Reset

[Duolingo lays off 10% of contractors amid AI push](#)

ADVANCED SEARCH OPTIONS ^

Date range: Custom 01/01/2024 01/31/2024

Location: Select country/territory

Industry: Education and training

AI principle: Select AI principle

Harm type: Economic/Property

Affected stakeholders: Workers

Business function: Select business function

AI system task: Select AI system tasks

AI system autonomy: Select AI system autonomy

Language: Select language

Type of search: Any of the concepts/keywords (OR)

Results: About 1 incidents & hazards

Number of results: 20

Sort by: Date of first reporting (most recent first)

Download results

 **Duolingo lays off 10% of contractors amid AI push**

2024-01-08 70 articles United States

Duolingo cut around 10% of its US-based contractors, replacing them with AI-driven content generation. Humans will still review AI outputs for accuracy, though former contractors report errors in lessons since the shift. Duolingo aims to optimize costs, reflecting a broader trend of AI displacing human labor. (AI GENERATED)

AI principles: Accountability Fairness Show More (3)

Industries: Education and training Consumer services

Affected stakeholders: Workers Consumers

Harm types: Economic/Property Reputational Psychological

Business function: Research and development Show More (1)

Autonomy level: Low-action autonomy (human-in-the-loop)

AI system task: Content generation

Why's our monitor labelling this an incident or hazard?

Figure 2. OECD AIM search result: “Duolingo lays off 10% of contractors amid AI push” (first reported 2024-01-08; 70 articles; United States). Search filters: 01/01/2024–01/31/2024; Education and training; Economic/Property; Workers. No autonomy filter was applied. Source: OECD (2024).

The record classifies the event under the values of Accountability and Fairness, among others. Those who were directly affected are Workers and Consumers. Types of harm include economic/property, reputational, and psychological. The AI task at hand is content generation. Autonomy Level is Low-action autonomy (human-in-the-loop).

There's a conflict point within the record. The record's summary is marked [AI GENERATED] and states that "Duolingo was replacing workers with AI-driven content generation." However, the same record also mentions the Low-action autonomy (human-in-the-loop) classification as only human reviewers are part of the process. Indeed, both statements can coexist: AI is decreasing the number of required employees but not fully removing human oversight. Still, the summary's use of "replacing" is a much bigger claim of causality than what Duolingo actually said (see below). The laboratory therefore is requesting to compare AIM with the reports from that company.

2.2 What happened

Question	Finding	Source
When did the reduction occur?	Contracts ended at the end of 2023. One contractor reports being offboarded on December 15, 2023. The news broke publicly on January 8, 2024.	Bloomberg (Ford & Lung, 2024); Campaign US (Doerr, 2024); TechCrunch (Forristal, 2024)
Who was affected?	About 10% of Duolingo's contractors. No full-time employees were affected.	Ford & Lung (2024); PYMNTS (2024)
What work was changing?	Translation and lesson-content creation. The CEO told shareholders in November 2023 that generative AI let Duolingo produce "new content dramatically faster," such as scripts for its language-learning shows.	Ford & Lung (2024); PYMNTS (2024)
What role did generative AI play?	A contributing cause, by Duolingo's own account. It is disputed whether AI was the <i>main</i> cause.	See Table 7
What did Duolingo say?	Its statements were inconsistent. They ranged from "part of that could be attributed to AI," to "not directly related to the use of AI," to "these are not layoffs."	Ford & Lung (2024); Capelouto (2024); Doerr (2024)
What is uncertain?	How many of the offboarded contractors were translators versus other roles. What share of the reduction AI caused. Whether lesson quality actually dropped. And whether the remaining contractors' work changed as described by former contractors.	—

2.3 Separating the evidence

Type	Content
Verified fact	Duolingo ended contracts for roughly 10% of its contractors at the end of 2023. Duolingo confirmed this to multiple outlets (Capelouto, 2024; Doerr, 2024; Ford & Lung, 2024). No full-time employees were affected (PYMNTS, 2024). The CEO's November 2023 shareholder letter credited generative AI with producing new content much faster. Duolingo had also launched Duolingo Max, an AI premium tier, in 2023 (PYMNTS, 2024).
Company statements	A spokesperson to Bloomberg said: "We just no longer need as many people to do the type of work some of these contractors were doing. Part of that could be attributed to AI" (Ford & Lung, 2024). To Semafor, Duolingo said humans will still review AI output, and that calling AI the sole reason would be an "oversimplification" (Capelouto, 2024). CMO Manu Orssaud told Campaign US the change was "not directly related to the use of AI" and that contracts end when "the job is complete." A spokesperson told PRWeek it affected a "small minority" and that "these are not layoffs" (Doerr, 2024).
Contractor claims	A Reddit user who said they had been a contract translator for five years wrote that Duolingo had offboarded "a huge percentage" of translation contractors. According to this user, two of their four-person team were let go, and the two who stayed would "just review AI content to make sure it's acceptable." They also said Duolingo's stated reason was that AI could produce content and translations. Users and critics reported mispronunciations and wrong translations after the shift. None of this has been independently verified (Albarino, 2024; Forristal, 2024).

Type	Content
My interpretation	Duolingo's three statements are not fully consistent. "Part of that could be attributed to AI" does not fit easily with "not directly related to the use of AI." Calling the cuts contract endings rather than layoffs is technically accurate, but it moves the event outside the category that normally triggers notice and consultation. Taken together, the most defensible reading is that AI changed <i>how much translation labor Duolingo needed</i> , even if other business reasons also played a part.

Table 7. Evidence separated by type, as the lab requires.

2.4 OECD record vs. contemporaneous reporting

The OECD record is accurate on the basics: the date, the roughly 10% figure, U.S. contractors, human review, and reports of errors. It differs from the primary reporting in three ways. (1) Its AI-generated summary says contractors were *replaced* by AI and its rationale says AI "directly" led to job cuts. Duolingo said only that AI was part of the reason. (2) It presents the contractors' error reports without noting they come from former contractors and users. (3) It files the case under "labor rights" harm, while Duolingo framed it as ordinary contract expiry. AIM is useful for *discovering* the incident, but its causal wording should not be cited as fact.

3. Step 3: Legal & Ethical Synthesis (Pillar 2 Alignment)

How Pillar 2 aligns with the theme: I interpret Pillar 2 as the trust and governance component of responsible AI: the legal and ethical safeguards, rights, transparency, human monitoring, and processes which make the use of AI trustworthy. This is the same description that the European Commission (2020) gives to the "ecosystem of trust", unlike the "ecosystem of excellence" that is about innovation and growth. The author in Chapter 17 does a similar thing regarding work: According to the European Social Partners, "Trustworthy AI" at work means lawful, ethical, and system-wide robust throughout the lifecycle (Ponce Del Castillo & Taes, 2025, p. 354). So this part takes the quantitative exposure from Step 1 and discusses its legal & ethical implications besides the case analysis and Chapter 17 labour law framework basis. Section 4 then turns that framework into the operating governance policy.

3.1 Displacement, augmentation, or restructuring?

A combination, with task restructuring as the mechanism. Duolingo reorganized its content pipeline so that AI produced the first draft. That restructuring produced two outcomes. For about 10% of contractors it meant **displacement**: their contracts ended. For those who stayed, if the contractor accounts are accurate, it meant **augmentation of a narrow kind**. They became reviewers of machine output rather than authors of translations. It was not pure augmentation, because some people lost their work. It was not pure displacement either, because humans remained in the loop, which is how the OECD record tags it.

The case is a clear example of **cognitive displacement**. Chapter 17 notes that AI adoption "initially" affected routine, lower-skilled work but "is extending to high-skilled workers who perform cognitive tasks" (Ponce Del Castillo & Taes, 2025, p. 342). Translation and lesson writing are knowledge work that depends on cultural and linguistic judgment. This is the same text-production category that scored highest on LLM exposure in my Business Analyst task analysis (Section 1.4).

3.2 Chapter 17 concerns applied to Duolingo

Chapter 17 concern	What the chapter says	Application to Duolingo
Worker autonomy and agency	Employers “must ensure that the use of AI systems respects workers’ agency”. Tacit knowledge built over years “should not be taken away from them and transferred to the machine” (p. 363).	Remaining translators reportedly moved from making judgments to checking a machine’s choices. Translators’ past work and corrections are the kind of expertise that improves AI output, so the knowledge transfer the chapter warns about is plausible here, though not documented.
Job content and skill requirements	Digital technology affects work content (skills), work organization, working conditions, and work relations. AI should “not jeopardize but augment human involvement and capacities at work” (European Social Partners, as cited in Ponce Del Castillo & Taes, 2025, pp. 353–354).	Work content shifted from producing translations to quality assurance. That is a narrower skill set, a sign of deskilling. It is the opposite of the “augment capacities” standard.
Employer power over restructuring	The employment relationship is defined by subordination and a power imbalance, and labor law exists to address the “unequal bargaining position” (p. 341).	Duolingo decided alone and let contracts lapse. Contractors are an especially weak group: ILO Recommendation 198’s indicators of employment (control, integration, continuity) may fit five-year “contractors”, but that status usually has to be proven (p. 340).
Transparency	What matters to workers is the system’s “intended purpose” and “context of use,” not access to the code (p. 363).	Duolingo gave inconsistent public explanations, and contractors learned the reason from an exit email. The purpose of the AI system (“new content dramatically faster”) was explained to shareholders before it was explained to the workers affected.
Consultation and social dialogue	ILO Recommendation 166: when technological change is “likely to entail terminations,” employers should consult workers’ representatives “as early as possible” (p. 350). AI risk assessments should be built <i>with</i> workers’ representatives (p. 364).	There is no record of consultation. The EU AI Act also would not have required any. Its worker-information duty (Art. 26(7)) covers only high-risk systems, such as recruitment and worker management, not content generation (pp. 356–357).

Table 8. Chapter 17 concerns applied to Duolingo. Page numbers in this table refer to Ponce Del Castillo and Taes (2025). ILO and European Social Partners instruments are cited as quoted in that chapter.

3.3 Is there a meaningful threat to dignity, agency, or expertise?

Professional expertise: yes, the most direct danger. One of the main reasons a translator would be useful is their ability to convey fine distinctions, use idioms correctly, and understand cultural context, all the above being forms of tacit knowledge. Such skill is developed mainly through years of experience, therefore it is neither easy to record nor suitable to provide to a machine. Limiting translator activity to just checking if AI output is “adequate” is lowering the bar from superb to just fair. Gradually, the number of people getting adequate exposure to acquire the skill through practice becomes smaller and smaller.

Worker agency: yes. Workers had no voice in the decision, and those who remained lost control over the core of their craft. Chapter 17 treats this loss of agency in human–machine work as a primary concern (Ponce Del Castillo & Taes, 2025, p. 363).

Human dignity: a moderate threat, not a severe one. Unlike the algorithmic firing (“robo-firing”) cases in the chapter (Ponce Del Castillo & Taes, 2025, pp. 345–346), no algorithm made decisions about individual people here. Humans made the business decision. The dignity concern is mainly about process: workers learned through an exit survey that they were replaceable, and the company’s public account minimized AI’s role.

Documented vs. my judgment. *Documented*: the contract reductions, their scale, the CEO's statement about faster content, the fact that AI was a partial reason, and continued human review. *Reported but unverified*: the translators' team composition, the reason contractors were given, the reviewers' changed role, and error reports. *My normative judgment*: that this was deskilling, that consultation should have happened, and that calling it a contract ending should not remove it from governance review.

3.4 What the legal framework means for the Step 1 risk scores

Chapter 17 helps turn the exposure scores from Step 1 into a judgment about *risk to workers*. Exposure alone is a technical measure. Legal and ethical risk is highest where three things meet: tasks with high LLM exposure, low accountability ratings, and workers with weak legal status. The Business Analyst tasks with the highest LLM score (4/5) are report writing, information gathering, and training materials. They carry lower accountability (3.00 on average for the most automatable tasks) than the planning and compliance work (4.00). That is the same profile as Duolingo's translation and lesson-writing work, which was done by contractors, the group with the least protection. By contrast, Medical Researcher combines low exposure with high complexity and near-zero adoption in a heavily regulated field, so its legal risk is low. Software Engineers have high exposure and high adoption, but most engineers are employees in a growing field. Its main risk is to entry-level work rather than to the occupation. Read this way, the Duolingo case is not an outlier. It is what the Step 1 data predict when exposed, lower-accountability tasks are held by workers outside labor law's protections (Ponce Del Castillo & Taes, 2025, pp. 340–341, 362–363).

4. Worker-Centric AI Policy

Purpose and scope. This policy governs any AI deployment that takes over, reshapes, or reduces demand for human tasks. “Workers” includes employees, contractors, and agency staff. Duolingo shows why: the displaced group was contractors, who sit furthest from consultation and redundancy protections (Ponce Del Castillo & Taes, 2025, pp. 340–341).

1. Material-change trigger

A deployment is a *material change* if it is expected to (a) automate 20% or more of a role’s working time, (b) reduce hours, contracts, or headcount, or (c) turn an expert role mainly into reviewing AI output. Working time is measured by a task-frequency audit: each task is weighted by hours workers log on it over a four-week baseline, and the worker panel validates the result. Management estimates alone cannot be used. Any trigger activates Sections 2–7.

2. Informing workers

Affected workers and their representatives receive written notice at least 60 days before deployment, stating which tasks AI will change or replace, why, and what happens to each role, including the system’s purpose and context of use (Ponce Del Castillo & Taes, 2025, p. 363). Whenever AI contributed to ending a contract, the exit notice must say so.

3. Displacement and deskilling assessment

Before deployment, a joint assessment identifies roles at risk of reduction, skills and tacit knowledge that will go unused, and quality risks. Task-level scores inform the assessment but never decide it, because they measure exposure, not outcomes.

4. Consultation

A worker panel that includes contractor representatives reviews the assessment “as early as possible” (ILO Recommendation No. 166, as cited in Ponce Del Castillo & Taes, 2025, p. 350) and before any irreversible decision. Management answers each objection in writing, and the panel helps set mitigation measures across the system’s life cycle (Ponce Del Castillo & Taes, 2025, p. 364).

5. Retraining and redeployment first

No workforce reduction caused wholly or partly by AI may proceed until the organization documents a search for redeployment options and offers paid retraining. Before ending any contract that has run continuously for 12 months or more, HR and legal assess misclassification risk against the ILO Recommendation 198 indicators, such as control, integration, and continuity (Ponce Del Castillo & Taes, 2025, p. 340). Where these indicate employment, employee protections apply. All other contractors receive a transition period or equivalent notice pay.

6. Human review, quality, and accountability

Human review is mandatory for AI output that (a) reaches customers or learners, (b) affects anyone’s pay, rights, or employment, or (c) belongs to a high-accountability task. Reviewers need domain expertise, authority to reject or rewrite output, and capped workloads. Each AI content pipeline has a named owner, and errors are traced to their source. At Duolingo, this would have given retained translators authority to rewrite AI output and a workload sized for linguistic nuance rather than volume.

7. Post-deployment monitoring

For 12 months after deployment, then annually, the organization reports to the worker panel on contract and headcount changes by role, error rates, reviewer workload, retraining outcomes, and user complaints. Rising errors or breached workload caps pause the deployment for joint review.

Ownership. The Chief AI Officer is accountable for compliance.

5. Policy Position

When does AI-enabled task automation become a labor-governance problem rather than merely a productivity improvement?

AI-enabled task automation becomes a labor-governance problem at the moment an organization turns the time AI saves into **less work, less pay, or a lower-skilled role**, and makes that choice **on its own**, without informing or consulting the people affected. The deciding factor is not how exposed a job is. It is what happens to the productivity gain, and who has a say in that decision. Three lines of evidence from this lab support that answer.

First, the exposure data show that a score cannot draw this line. Software Engineer has the highest contact level based on all metrics with 28.4% observed adoption; employment for the job, however, is expected to increase about 16% at the same time where AI is speeding up processes and engineers are still making the decisions, owning the work, and keeping their jobs. That is a productivity improvement. Business Analyst has almost the same Automation score (3.09 vs. 3.10) but 22 times lower adoption, and its most exposed tasks are text production: reports, information gathering, and training materials. Two roles with nearly identical scores are living very different realities. Whether exposure becomes acceleration or displacement is decided by the organization, not by the score.

Second, Duolingo shows what happens when that decision goes the other way. Its exposure was the same kind the Business Analyst task list reveals: generating text and translations. Duolingo converted the gain (“new content dramatically faster”) into about 10% fewer contractors (Ford & Lung, 2024). According to contractor accounts, the translators who stayed were reassigned to checking AI output (Forristal, 2024). The decision was unilateral, the public explanation was inconsistent, and the people affected were contractors, outside the consultation rights that labor law normally provides. No exposure score predicted this outcome. The decision produced it.

Third, Chapter 17 explains why that decision is a matter of governance rather than efficiency. Labor law exists because the employment relationship is unequal (Ponce Del Castillo & Taes, 2025, p. 341). A choice about who keeps their work, and on what terms, is a choice about power. The chapter also argues that voluntary ethics guidelines “cannot sufficiently guarantee adequate workers’ protection” and that enforceable rules are needed (Ponce Del Castillo & Taes, 2025, p. 354). Yet the Duolingo case falls through the gaps in the rules that do exist, on both sides of the Atlantic. Article 26(7) of the EU AI Act limits the duty to inform workers to high-risk systems, meaning the recruitment and worker-management tools listed in Annex III (Ponce Del Castillo & Taes, 2025, pp. 356–357). A generative content tool that shrinks a contractor pool is not on that list. The contractors themselves also fall outside most information, consultation, and redundancy rights, which attach to employees, unless they can first prove an employment relationship. In the United States, no federal rule required notice or consultation for contractors. Where statutory labor law does not reach non-standard workers in either jurisdiction, a company-wide policy that applies one standard everywhere the company operates is the only instrument that can close the gap.

Conclusion. Exposure scores tell us *where* automation pressure will land. Governance decides *what that pressure does to people* when it lands. Automation becomes a labor-governance problem when three conditions occur together: the gains are taken out as reduced work instead of being shared, workers’ expertise is downgraded to supervising the machine, and the people affected have no voice before the decision is made. My Worker-Centric AI Policy is designed to break that chain at each link. The notice and consultation clauses restore voice. The deskilling assessment and human-review standards protect expertise. The redeployment-first rule, which extends to contractors through the misclassification check, keeps gains from being taken only as cuts.

References

AI Labor Index. (n.d.-a). *Business & management consultant*. Retrieved September 24, 2026, from <https://www.ailaborindex.com/occupations/business-management-consultant>

AI Labor Index. (n.d.-b). *Medical researcher*. Retrieved September 24, 2026, from <https://www.ailaborindex.com/occupations/medical-researcher>

AI Labor Index. (n.d.-c). *Software engineer*. Retrieved September 24, 2026, from <https://www.ailaborindex.com/occupations/software-engineer>

Albarino, S. (2024, January 9). Duolingo translator layoffs spark AI debate. *Slator*. <https://slator.com/duolingo-translator-layoffs-spark-ai-debate/>

Capelouto, J. D. (2024, January 9). Duolingo cuts contractors as it further embraces AI. *Semafor*. <https://www.semafor.com/article/01/09/2024/duolingo-cuts-contractors-as-it-further-embraces-ai>

Doerrer, B. (2024, January 9). Duolingo offboards translation contractors; workers allege AI replacement. *Campaign US*. <https://www.campaignlive.com/article/duolingo-offboards-translation-contractors-workers-allege-ai-replacement/1856925>

European Commission. (2020). *White paper on artificial intelligence: A European approach to excellence and trust* (COM(2020) 65 final). <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020DC0065>

Ford, B., & Lung, N. (2024, January 8). Duolingo cuts 10% of contractors as it uses more AI to create app content. *Bloomberg*. <https://www.bloomberg.com/news/articles/2024-01-08/duolingo-cuts-10-of-contractors-in-move-to-greater-use-of-ai>

Forristal, L. (2024, January 9). Duolingo cut 10% of its contractor workforce as the company embraces AI. *TechCrunch*. <https://techcrunch.com/2024/01/09/duolingo-cut-10-of-its-contractor-workforce-as-the-company-embraces-ai>

OECD. (2024, January 8). *Duolingo lays off 10% of contractors amid AI push* [Incident record]. OECD AI Incidents and Hazards Monitor. Retrieved September 24, 2026, from <https://oecd.ai/en/incidents/2024-01-08-13c5>

Ponce Del Castillo, A., & Taes, S. (2025). Artificial intelligence and labor law. In N. A. Smuha (Ed.), *The Cambridge handbook of the law, ethics and policy of artificial intelligence* (pp. 339–366). Cambridge University Press. <https://doi.org/10.1017/9781009367783.021>

PYMNTS. (2024, January 8). Duolingo cuts 10% of contractors while expanding use of AI. <https://www.pymnts.com/news/artificial-intelligence/2024/duolingo-cuts-10percent-contractors-expanding-use-of-ai/>

TripleTen. (n.d.). *Will AI take my job? Free AI job risk calculator* [Online calculator]. Retrieved September 24, 2026, from <https://tripleten.com/tools/what-jobs-will-ai-replace/>

AI Assistance: Used Grammarly for minor editing only.

Appendix: Screenshots

A1. Software Engineer

Will AI take my job?

Find out in 30 seconds. Our free calculator shows you what to do next.

Job title

Software Engineer

Industry

Technology & Engineering

Company size

Small

Medium

Enterprise

Experience

Early

Mid

Senior

Remote

Do you work remotely?

Continue

45%

exposure

Moderate exposure

Writing the code is the part models got good at first, so the share of your week spent typing known patterns is genuinely exposed. Employment tells a different story: BLS projects software developer roles to grow about 16% through 2034. What holds is deciding what to build, catching the design that will hurt in six months, and owning the thing once it's running. Five years of context is the part that transfers worst to a tool, and it pulls your number down.

You answered like someone who runs inference in production. There is little gap here. The open question is whether your title and pay have caught up.

This score is task exposure, not a probability of losing your job. It combines measured AI use by occupation with US Bureau of Labor Statistics employment projections through 2034.

TripleTen inputs (Technology & Engineering; Small; Senior; not remote) and result: 45%, Moderate exposure.

Will AI take my job?

Find out in 30 seconds. Our free calculator shows you what to do next.

Job title

Software Engineer

Industry

Technology & Engineering

Company size

Small

Medium

Enterprise

Experience

Early

Mid

Senior

Remote

Do you work remotely?

Continue

53%

exposure

Moderate exposure

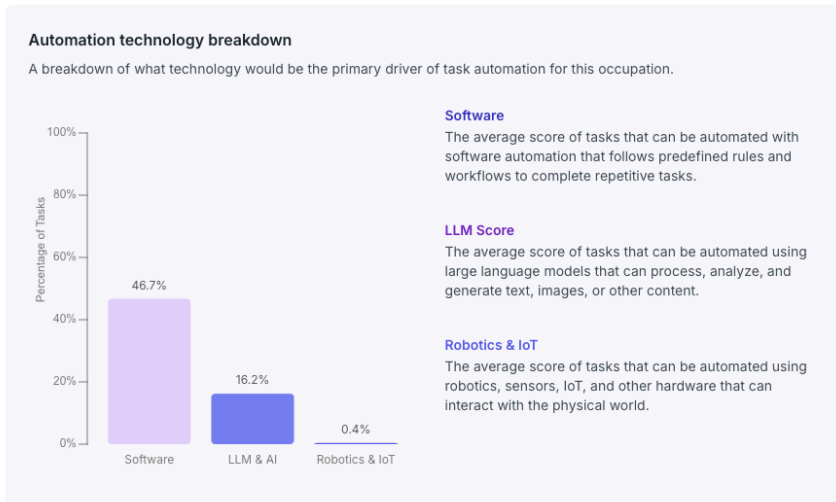
Writing the code is the part models got good at first, so the share of your week spent typing known patterns is genuinely exposed. Employment tells a different story: BLS projects software developer roles to grow about 16% through 2034. What holds is deciding what to build, catching the design that will hurt in six months, and owning the thing once it's running. Tech companies adopt these tools first, so you feel the change before other sectors do.

You answered like someone who runs inference in production. There is little gap here. The open question is whether your title and pay have caught up.

This score is task exposure, not a probability of losing your job. It combines measured AI use by occupation with US Bureau of Labor Statistics employment projections through 2034.

Controlled baseline rerun: TripleTen inputs (Technology & Engineering; Medium; Mid; not remote) and result: 53%, Moderate exposure.

Role	Complexity	Automation ↓	Adoption	Economic impact
3 Software Engineer Develops, tests, and maintains software applications, ensuring functionality, security, and efficiency.	3.71	3.10	28.4%	\$288.17B >



AI Labor Index: Complexity 3.71, Automation 3.10, Adoption 28.4%. Technology breakdown: Software 46.7%, LLM & AI 16.2%, Robotics & IoT 0.4%.

A2. Business Analyst (AI Labor Index: Business & Management Consultant)

Will AI take my job?

Find out in 30 seconds. Our free calculator shows you what to do next.

Job title
Business Analyst

Industry
Finance & Accounting

Company size
Small (1-50) Medium (51-500) Enterprise (500+)

Experience
Early (0-2 years) Mid (3-5 years) Senior (5+ years)

Remote
Do you work remotely? ☐

Continue

29% exposure

Lower exposure

Reporting, planning documents, and status-chasing are the parts a model handles well, and they take up more of the week than most managers admit. Deciding, prioritizing, and being accountable for the outcome don't move. Consulting and operations analysis are both projected to keep growing, so the pressure is on how the work gets done. Large companies run specialized, process-heavy work, and that is what automation budgets go looking for.

You think about AI the way people who work alongside it do. That is an advantage in a role like yours, and it is worth making visible.

This score is task exposure, not a probability of losing your job. It combines measured AI use by occupation with US Bureau of Labor Statistics employment projections through 2034.

TripleTen inputs (Finance & Accounting; Enterprise; Mid; remote) and result: 29%, Lower exposure.

Will AI take my job?

Find out in 30 seconds. Our free calculator shows you what to do next.

Industry

Finance & Accounting

Company size

Small1-50

Medium51-500

Enterprise500+

Experience

Early0-2 years

Mid3-5 years

Senior5+ years

Remote

Do you work remotely?

Continue

Will AI take my job?

Find out in 30 seconds. Our free calculator shows you what to do next.

25%
exposure

Lower exposure

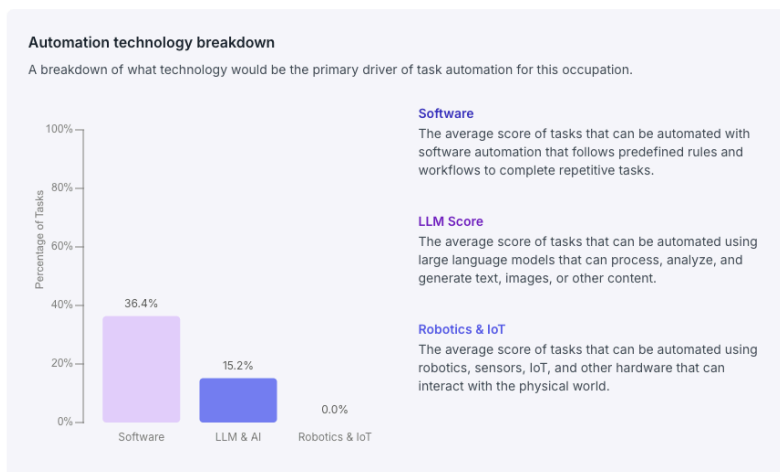
Reporting, planning documents, and status-chasing are the parts a model handles well, and they take up more of the week than most managers admit. Deciding, prioritizing, and being accountable for the outcome don't move. Consulting and operations analysis are both projected to keep growing, so the pressure is on how the work gets done. Finance has been automating documented processes for decades, and it has the budget to keep going.

You think about AI the way people who work alongside it do. That is an advantage in a role like yours, and it is worth making visible.

This score is task exposure, not a probability of losing your job. It combines measured AI use by occupation with US Bureau of Labor Statistics employment projections through 2034.

Controlled baseline rerun: TripleTen inputs (Finance & Accounting; Medium; Mid; not remote) and result: 25%, Lower exposure.

Role	Complexity	Automation ↓	Adoption	Economic impact
Business & Management Consultant				
5 Provides strategic advice to businesses on efficiency, growth, and market competition.	4.03	3.09	1.3%	\$561.76B >



AI Labor Index: Complexity 4.03, Automation 3.09, Adoption 1.3%. Technology breakdown: Software 36.4%, LLM & AI 15.2%, Robotics & IoT 0.0%.

A3. Medical Researcher

Will AI take my job?

Find out in 30 seconds. Our free calculator shows you what to do next.

Job title

Medical Researcher

Industry

Healthcare & Medical

Company size

Small
1-50

Medium
51-500

Enterprise
500+

Experience

Early
0-2 years

Mid
3-5 years

Senior
5+ years

Remote

Do you work remotely?

☐

Continue

19%

exposure

Lower exposure

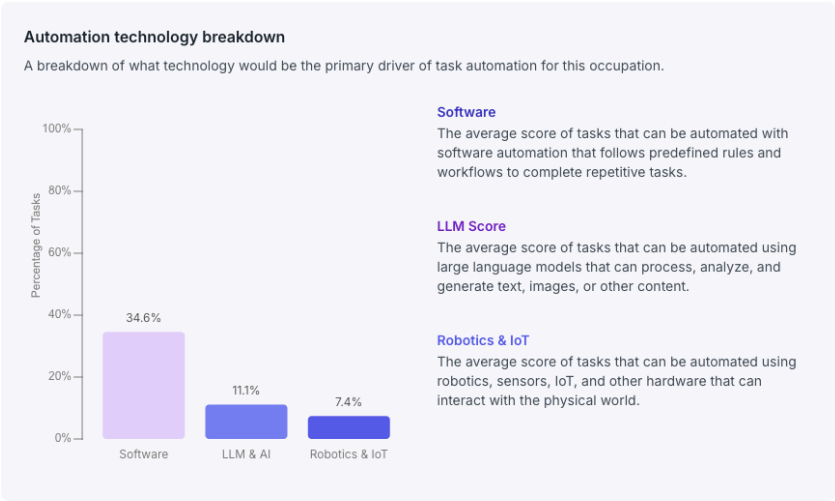
Licensing, physical presence, and legal responsibility for the outcome keep this work firmly with people. AI arrives here as an instrument, in imaging, notes, and decision support, and healthcare employment keeps growing for reasons that have nothing to do with technology. Expect pressure on how you work rather than on whether the job exists. Healthcare moves slowly on anything regulated, and that buys the whole sector time.

You think about AI the way people who work alongside it do. That is an advantage in a role like yours, and it is worth making visible.

This score is task exposure, not a probability of losing your job. It combines measured AI use by occupation with US Bureau of Labor Statistics employment projections through 2034.

TripleTen inputs (Healthcare & Medical; Medium; Mid; not remote) and result: 19%, Lower exposure.

Role	Complexity	Automation ↓	Adoption	Economic impact
15 Medical Researcher Conducts scientific research to advance medical knowledge, treatments, and public health.	4.09	2.90	0.3%	\$85.53B >



AI Labor Index: Complexity 4.09, Automation 2.90, Adoption 0.3%. Technology breakdown: Software 34.6%, LLM & AI 11.1%, Robotics & IoT 7.4%.

A4. Business & Management Consultant task list (AI Labor Index)

Tasks involved

This is the full list of tasks and their labeled scores for this occupation. The dataset for this occupation consists of 66 tasks and 4 suboccupations.

Task	Occupation	Complexity	Automation +	LLM	Occupation	Complexity	Automation +	LLM & AI	Accountability	Task nature
Establish, maintain, or test call trees to ensure appropriate communication during disaster.	Business Continuity Planners	3/5	4/5		Business Continuity Planners	3/5	4/5	3/5	3/5	Digital ✎
Prepare reports summarizing operational results, financial performance, or accomplishments of specified objectives, goals, or plans.	Business Continuity Planners	3/5	4/5		Business Continuity Planners	3/5	4/5	4/5	3/5	Digital ✎
Analyze corporate intelligence data to identify trends, patterns, or warnings indicating threats to security of people, assets, information, or infrastructure.	Business Continuity Planners	4/5	4/5		Business Continuity Planners	4/5	4/5	4/5	4/5	Digital ✎
Study and analyze information about alternative courses of action to determine which plan will offer the best outcomes.	Operations Research Analysts	4/5	4/5		Operations Research Analysts	4/5	4/5	4/5	3/5	Digital ✎
Develop and apply time and cost networks to plan, control, and review large projects.	Operations Research Analysts	4/5	4/5		Operations Research Analysts	4/5	4/5	3/5	3/5	Digital ✎
Prepare staff work schedules and assign specific duties.	General and Operations Managers	3/5	4/5		General and Operations Managers	3/5	4/5	3/5	2/5	Digital ✎
Review financial statements, sales or activity reports, or other performance data to measure productivity or goal achievement or to identify areas needing cost reduction or program improvement.	General and Operations Managers	4/5	4/5		General and Operations Managers	4/5	4/5	3/5	3/5	Digital ✎
Gather and organize information on problems or procedures.	Management Analysts	3/5	4/5		Management Analysts	3/5	4/5	4/5	3/5	Digital ✎
Develop emergency management plans for recovery decision making and communications, continuity of critical departmental processes, or temporary shut-down of non-critical departments to ensure continuity of operation and governance.	Business Continuity Planners	5/5	3/5		Business Continuity Planners	5/5	3/5	3/5	4/5	Digital ✎

Tasks involved

This is the full list of tasks and their labeled scores for this occupation. The dataset for this c 66 tasks and 4 suboccupations.

Task	Occupation	Complexity
Develop disaster recovery plans for physical locations with critical assets, such as data centers.	Business Continuity Planners	4/5
Write reports to summarize testing activities, including descriptions of goals, planning, scheduling, execution, results, analysis, conclusions, and recommendations.	Business Continuity Planners	3/5
Analyze impact on, and risk to, essential business functions or information systems to identify acceptable recovery time periods and resource requirements.	Business Continuity Planners	4/5
Review existing disaster recovery, crisis management, or business continuity plans.	Business Continuity Planners	4/5
Conduct or oversee contingency plan integration and operation.	Business Continuity Planners	4/5
Interpret government regulations and applicable codes to ensure compliance.	Business Continuity Planners	5/5
Create scenarios to reestablish operations from various types of business disruptions.	Business Continuity Planners	5/5
Create or administer training and awareness presentations or materials.	Business Continuity Planners	3/5
Identify opportunities for strategic improvement or mitigation of business interruption and other risks caused by business, regulatory, or industry-specific change initiatives.	Business Continuity Planners	5/5
Create business continuity and disaster recovery budgets.	Business Continuity Planners	4/5

Tasks involved

This is the full list of tasks and their labeled scores for this occupation. The dataset for this occupation consists of 66 tasks and 4 suboccupations.

	Occupation	Complexity	Automation +	LLM & AI	Accountability
ry plans for critical iters.	Business Continuity Planners	4/5	3/5	3/5	4/5
ize testing rscriptions of ling, sis, mendations.	Business Continuity Planners	3/5	3/5	4/5	3/5
risk to, ions or identify e periods nts.	Business Continuity Planners	4/5	3/5	3/5	3/5
recovery, usiness	Business Continuity Planners	4/5	3/5	3/5	3/5
tingency ration.	Business Continuity Planners	4/5	3/5	3/5	4/5
gulations ensure	Business Continuity Planners	5/5	3/5	3/5	4/5
tablish types of	Business Continuity Planners	5/5	3/5	3/5	4/5
ning and s or	Business Continuity Planners	3/5	3/5	4/5	3/5
r strategic on of d other risks ulatory, or e initiatives.	Business Continuity Planners	5/5	3/5	3/5	4/5
ity and ts.	Business Continuity Planners	4/5	3/5	3/5	3/5

Tasks involved: 66 tasks across 4 sub-occupations, sorted by Automation (descending). Each pair shows the left and right halves of the same table.